Executive Summary

This report analyzes at a sample of 100 used Volkswagen Golf listings within 10 miles of Birmingham (B4 7ET) from the larger UK market on AutoTrader (2025) to find the most important things that affect price. The results show that the local prices (£15,903) are very close to the national benchmark (£15,655), which means that Birmingham's market (B4 7ET) reflects pricing trends in the rest of the UK. The price goes down the most when the car is older and has more miles on it. Bigger engines and automatic transmissions, on the other hand, make the price go up. A Chi-Square test showed a strong link between fuel type and efficiency, with diesel and hybrid models always getting better gas mileage. The regression model accounted for 88% of price variation, validating that these quantifiable attributes predominantly influence market disparities. There was some minor non-linearity because there wasn't enough hybrid data, but the overall model is still strong. It is recommended that key focus should be on mid-priced (£10K–£20K) petrol automatics, as they dominate demand and yield higher margins, while diesels serve the low-cost segment and hybrids the eco-premium tier. Highlight high-spec black automatics with 4WD or 2.0L engines as premium “hero cars” to attract faster, higher-value sales.

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1. Introduction

This report was prepared for Car4All, a market research aimed at analyzing the second-hand car market in Birmingham (B4 7ET), focusing specifically on used Volkswagen Golf models. The goal is to find the main things that affect car costs and see if prices in the area are different from those across the country. Utilizing data from AutoTrader.co.uk (2025), a random sample of 100 listings was examined employing descriptive statistics, correlation analysis, hypothesis testing, and regression modeling. The results give Car4All evidence-based ideas for how to improve their pricing tactics, understand how the market works, and make a car price calculator that can be used in other regions as well.

2. Sampling Method

A random sample of 100 used Volkswagen Golf cars was selected from a population of 591 listings on AutoTrader.uk within a 10-mile radius of postcode B4 7ET, Birmingham. The simple random sampling method ensured each car had an equal chance of selection, improving representativeness and reducing bias. The 10-mile range captured local market variation, and the 100-sample size provided sufficient diversity for reliable, regionally relevant price analysis. Data were collected on 10 December 2025.

3. Limitations

There were certain limits to the analysis. The dataset only has 100 listings, which may not show all the different ways that used Volkswagen Golfs are sold in the UK. There weren't enough hybrid and plug-in hybrid cars in the sample, which could make comparisons of price and efficiency a little unfair. Also, time limits made it hard to gather long-term data. Even with these problems, the sample gives a strong picture of how the local market works and gives us solid information about what affects used car pricing in Birmingham.

4. Descriptive Analysis: Exploring Patterns and Determinants of VW Golf Prices

4.1 Population - Market Distribution and Central Tendency of Golf Prices

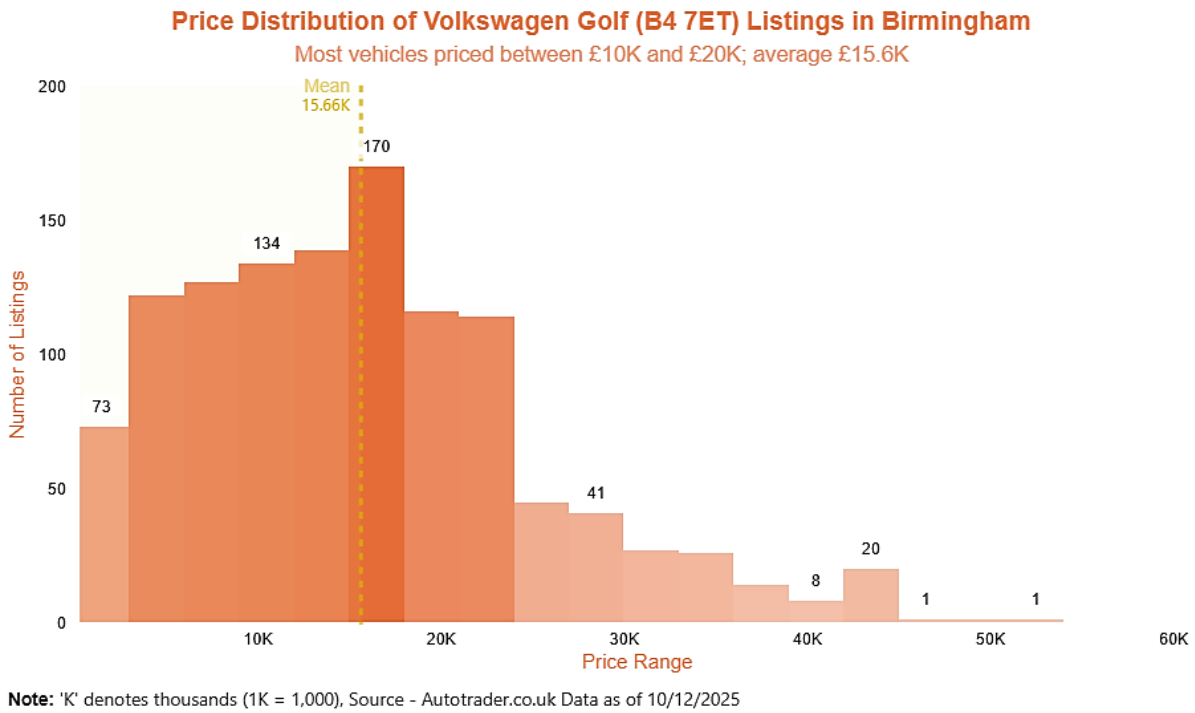


Figure 1: Price Distribution of Population

Figure 1 illustrates the price distribution of second-hand Volkswagen Golf (B4 7ET) vehicles listed in Birmingham. The majority of listings fall between £10,000 and £20,000, with the average price calculated at £15.66K. The right-skewed shape of the distribution suggests that while most vehicles are mid-priced, a smaller number of premium listings elevate the average. Stakeholders should focus their inventory and pricing strategy around the £10K–£20K range for faster turnover, while selectively promoting hybrid listings above £25K as a premium, future-oriented offering to diversify revenue streams

Figure 1 follows Tufte's principles of Graphical Integrity and Graphical Excellence, showing prices accurately with equal bin widths and undistorted scaling (Simpson, 2025). The price bands (£0–£10K, £10–£20K, etc.) follow IBCS guidelines, helping viewers easily identify where most listings fall (IBCS, 2022).

Based on Gestalt principles, the gradient orange color groups the bars visually and highlights the busiest range (£10K–£20K), making the key message stand out (Simpson, 2025). Only key bars were labelled to keep the graph clean in line with Tufte's Data-Ink Ratio and IBCS clarity standards. Adding labels to all bins would cause clutter, so selective labelling helps viewers focus on the most relevant information. Overall, the design balances honesty, simplicity, and readability, clearly showing that most VW Golf listings are mid-priced, around £15.6K (IBCS, 2022).

4.2 Sample - Distribution of Used VW Golf Listings by Price Range and Fuel Type

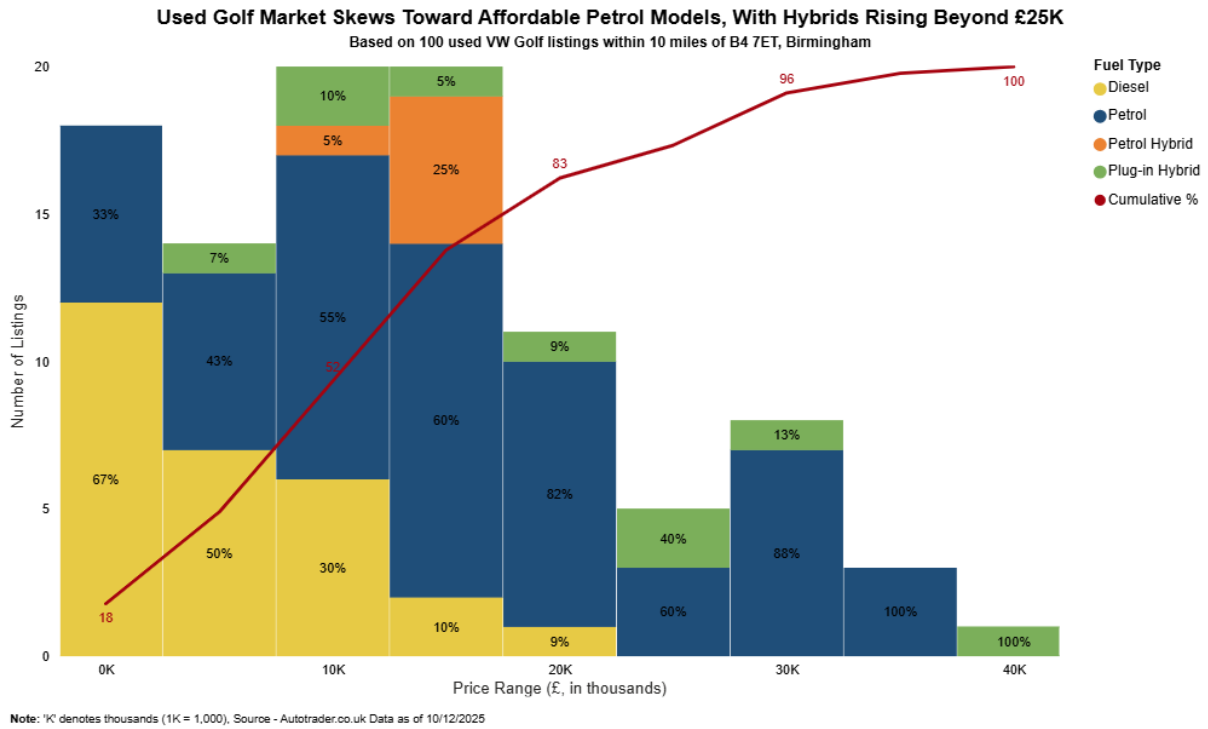


Figure 2: Sample Price Distribution by Fuel Type

Figure 2 shows how the prices of 100 used Volkswagen Golf cars within 10 miles of Birmingham (B4 7ET) are spread out, according on the type of fuel they use. The market is very focused on inexpensive models, since around 80% of listings are priced below £20K. Diesel models make up most of the lower bands since they are older, whereas hybrids and plug-in hybrids show up after £25K, which means they are newer, higher-spec. This gives a clear picture of how the type of fuel affects pricing differences in the local market.

To keep things accurate, the graphic follows Tufte's rules for Graphical Integrity and Excellence by using equal bin widths and undistorted scaling (Simpson, 2025). The color scheme, diesel in yellow, petrol in blue, and hybrids in green and orange, follows Gestalt principles to make it easier to see patterns. The red cumulative line leverages figure-ground contrast to show where the market is concentrated. To keep IBCS clarity and Data-Ink Ratio criteria, labels are only used in certain places (IBCS, 2022). Using different colors in stacked bars was a deliberate violation of IBCS's monochromatic guidance to make it easier for stakeholders to understand. (IBCS, 2022).

4.3 Impact of Mileage on Price Depreciation

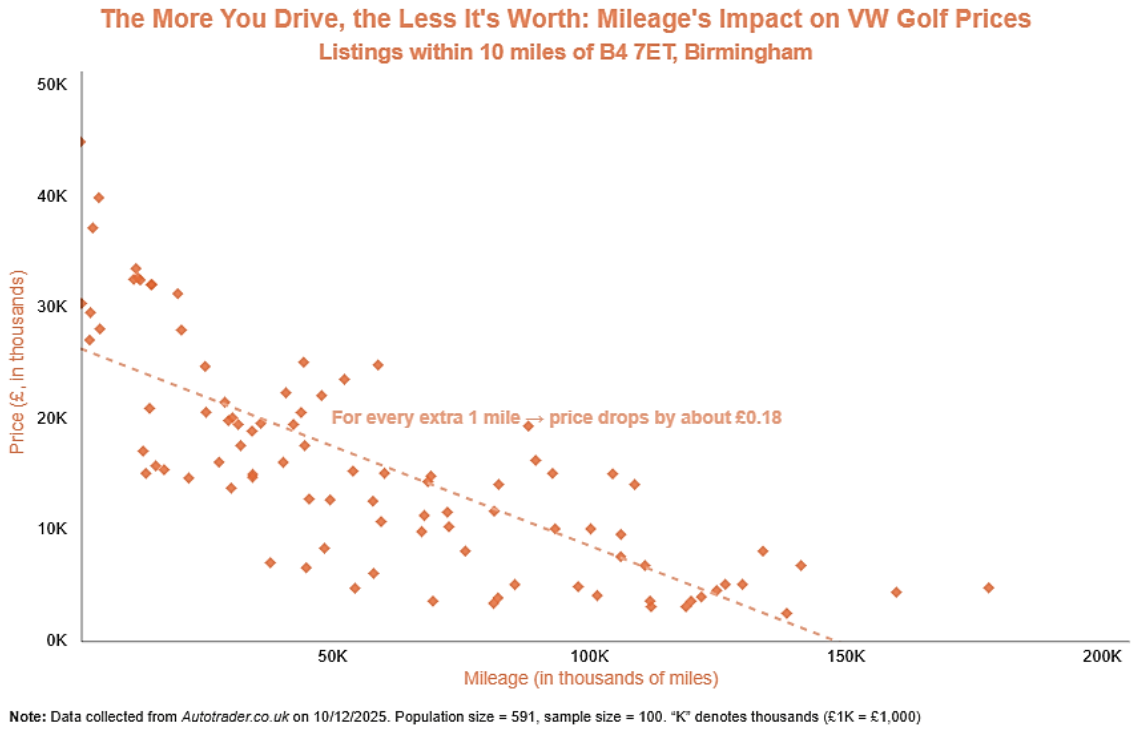


Figure 3: Mileage VS Price

Figure 3 makes it evident that there is a strong negative association between the price of a VW Golf and the number of miles it has driven. It shows that the worth of an automobile goes down as the miles goes up, which is an example of how wear and tear can lower its value. The regression line shows this relationship in a straightforward way: on average, each extra mile lowers the price by roughly £0.18.

To follow Tufte's concepts of Graphical Integrity and Graphical Excellence, a continuous and proportional scaling of the axes keeps the trend from being exaggerated, which keeps the data's true relationship intact. Following Gestalt's concept of resemblance, a single orange hue was used to visually bring the data points together while keeping the focus on the regression line (Simpson, 2025). The annotation ("For every extra 1 mile → price drops by about £0.18") is a pre-attentive signal that helps non-technical viewers understand the slope right away, which is in line with IBCS's criteria for clear and message-first communication (IBCS, 2022).

However, the design goes against Tufte's Data-Ink Ratio principle on purpose by adding an annotation and color emphasis, which add ink that isn't needed. This trade-off was deliberate: the annotation and tone gradient make it easier for people who aren't used to statistical charts to understand them (IBCS, 2022).

4.3 Influence of Engine, Ownership, Age, and Color on Valuation

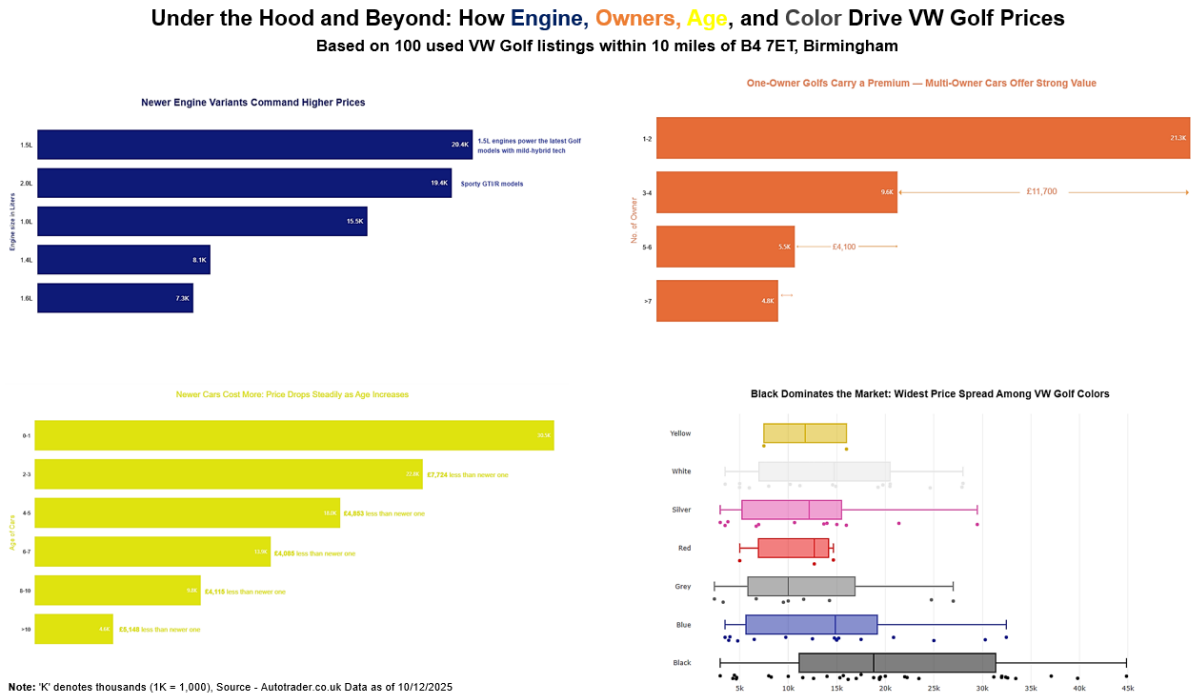


Figure 4: Influence of Different Variable on Price

This multi-chart visual shows how engine type, ownership history, age, and color influence VW Golf prices within 10 miles of B4 7ET, Birmingham. It reveals newer 1.5L and sporty 2.0L engines command the highest values due to mild-hybrid and sports technology, while older 1.4L and 1.6L variants are considerably cheaper. Cars with a single owner are priced up to £11.7K higher, reflecting buyer preference for well-maintained vehicles. Vehicle age also shows steady depreciation, and black Golfs display the widest price range, suggesting broad appeal across both economy and premium segments.

From a design standpoint, Figure 4 follows Tufte's principles of Graphical Integrity and Excellence, ensuring that price differences are shown accurately and proportionally (Simpson, 2025). Gestalt principles of proximity and similarity guided the layout — related charts are grouped together and share consistent formatting, allowing easy visual comparison. Annotations such as "£7,724 less than newer one" is for helping stakeholders interpret key insights instantly. The color scheme adheres to IBCS clarity standards, with distinct hues used to represent each factor while maintaining uniformity within each chart (IBCS, 2022). The silver category was intentionally colored pink to improve contrast and visibility against the white background, maintaining interpretive accuracy. Although the use of annotations and bright colors slightly reduces Tufte's data-ink ratio, this trade-off enhances comprehension, ensuring even non-technical audiences can extract the intended insights with ease.

5. DESCRIPTIVE STATISTICS

Variable	Category	Count	Mean	Median	Standard Deviation	Minimum Value	25%	50%	75%	Maximum Value
Price	Diesel	28	**8178.29**	5495	5281.238	2990	4070	5495	12268.75	24621
	Petrol	57	18445.37	15987	9728.931	2395	12091.5	15987	24874.5	39814
	Petrol Hybrid	6	17949.83	19399.5	3315.428	11202	16899.75	19399.5	19424.75	19499
	Plug-in Hybrid	9	**22476.89**	20499	11744.473	7990	12595	20499	29884	44845
	Automatic	54	19801.04	19099	10397.522	2990	10536.25	19099	28368.25	44845
	Manual	46	11328.54	12099.5	6916.115	2395	4758.75	12099.5	15987	27000
Mileage	Diesel	28	**89710.04**	86045	37789.516	25302	58030.75	86045	119750	178000
	Petrol	57	48956.51	40511	36718.653	2780	15304.5	40511	82006	141424
	Petrol Hybrid	6	42557.5	39327.5	13201.494	31738	33765.25	39327.5	48883.25	68000
	Plug-in Hybrid	9	**31551.56**	13750	44655.383	1000	2072	13750	50401	134000
	Automatic	54	53388.7	43255.5	42948.079	1000	14889	43255.5	86667.5	178000
	Manual	46	64320.02	56312	39631.89	2780	30562.75	56312	90712.25	141424
Age of cars	Diesel	28	**9.3**	9	3.6	3	6.5	9	12	16
	Petrol	57	5.7	4	4.9	1	2	4	8	17
	Petrol Hybrid	6	4	4.5	1.7	2	2	4.5	5	6
	Plug-in Hybrid	9	**3.6**	2	3.7	1	1	2	7	9
	Automatic	54	5.5	5	4.1	1	2	5	8	16
	Manual	46	7.5	6	5.2	1	3	6	12	16

Table 1

Table 1 summarizes car price, mileage, and age by fuel and transmission type. Mean, standard deviation, and quartiles show group differences.

Diesel automobiles had the lowest average price but the highest average mileage and oldest age, showing they are older, depreciated vehicles. Their wider quartile range includes older and heavily used models. Petrol and automatic cars are mid-priced, reflecting balanced use and age.

Plug-in hybrids have the highest average costs but the lowest mileage and youngest age, indicating that they are newer, more advanced, and worth more. Petrol hybrids' high costs and little variability indicate ongoing demand for efficient, traditional solutions.

A 95% confidence interval was made for both front-wheel drive and four-wheel drive cars to identify whether price is influenced by drive type or not. Table 2 shows that the real average price for Four-Wheel Drive cars is between £24,169.39 and £33,451.89, and for Front-Wheel Drive cars it is between £11,984.29 and £15,620.82. This is based on the idea that car prices are normally-distributed.

	Four Wheel Drive (£)	Front Wheel Drive (£)
Mean Price	28810.64	13802.56
Lower Bound	24169.39	11984.29
Upper Bound	33451.89	15620.82

Table 2

Four-Wheel Drive cars have a much wider confidence interval, which means they are more expensive. This is probably because their engine systems are more complex and they can be used for high-end or off-road models. A 95% confidence level was picked because it is a good mix between accuracy and dependability. It can be used to look at changes in market prices instead of making exact pricing decisions.

6. HYPOTHESIS TESTING

6.1 One-Sample T-Test: Comparing Local Market Prices with National Benchmarks

Research Question Hypothesis,

- Do Volkswagen Golf listings in the Birmingham (B4 7ET) sample have an average price that is statistically different from the national benchmark of £15,655.22?

1. Null Hypothesis (H_0): The mean price of Volkswagen Golf cars in the Birmingham (B4 7ET) sample is equal to the national mean price (£15,655.22).
2. Alternative Hypothesis (H_1): The mean price of Volkswagen Golf cars in the Birmingham (B4 7ET) sample is not equal to the national mean price (£15,655.22).
3. Statistical Test: One-Sample t-test
4. Significance Level: 5% (two-tailed test, α = 0.05)
5. Calculated Test Statistic: t(99) = 0.25, p = 0.802
6. Technical Conclusion: Since p > 0.05, fail to reject the null hypothesis.
7. Interpretation: There is no statistically significant difference between the average price of Volkswagen Golf cars in the Birmingham (B4 7ET) sample (£15,903.69) and the national benchmark of £15,655.22. The 95% confidence interval for the mean difference (−£1,712.09 to £2,209.03). This suggests that local market prices align closely with national trends, implying that used Volkswagen Golf prices in Birmingham are representative of the broader UK market without any meaningful regional price deviation.

T-Test

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
Price	100	15903.69	9880.760	988.076

Table 3

One-Sample Test

Test Value = 15655.22

	t	df	Significance One-Sided p	Significance Two-Sided p	Mean Difference	95% Confidence Interval of the Difference Lower	95% Confidence Interval of the Difference Upper
Price	.251	99	.401	.802	248.470	-1712.09	2209.03

Table 4

6.2 Chi-Square Test: Relationship Between Fuel Type and Efficiency Classification

A Chi-Square test of independence was conducted to examine the relationship between fuel type and fuel efficiency category. Used ranges to align with UK industry benchmarks, where vehicles achieving over 50 mpg are regarded as highly efficient. This categorization allowed the grouping of vehicles into clear and comparable classes for analysis. (Cinch, 2025) (Carwow, 2025)

Category	**MPG Range**
Low Efficiency	≤ 35 mpg
Moderate Efficiency	36 – 50 mpg
High Efficiency	≥ 51 mpg

Table 5

Research Question Hypothesis

- Is there a significant association between fuel type and fuel efficiency among Volkswagen Golf cars in the sample?

1. Null Hypothesis (H_0): Fuel type and fuel efficiency are not associated of each other.
2. Alternative Hypothesis (H_1): Fuel type and fuel efficiency are associated with each other.
3. Statistical Test: Chi-Square Test of Independence
4. Significance Level: 5% ($\alpha = 0.05$)
5. Calculated Test Statistic: $\chi^2(6, N = 100) = 60.797$, $p < 0.05$
6. Technical Conclusion: Since $p < 0.05$, reject the null hypothesis of independence.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	60.797[a]	6	<.001
Likelihood Ratio	71.792	6	<.001
N of Valid Cases	100		

a. 6 cells (50.0%) have expected count less than 5. The minimum expected count is 1.20.

Table 6

7. Interpretation: The test results indicate a statistically significant relationship between fuel type and fuel efficiency. The Cramer's V value of 0.551(Table 7) exceeds the 0.5 threshold, signifying a strong association between these two variables. This suggests that fuel type meaningfully influences a car's fuel efficiency. Diesel and plug-in hybrid cars dominate the high-efficiency range, while petrol vehicles are mostly found in the moderate efficiency group (Figure 5). These outcomes align with established automotive patterns, diesel and hybrid engines are generally more fuel-optimized due to their advanced combustion and energy recovery systems. However, careful interpretation is needed, as the limited number of hybrid vehicles in the dataset may amplify their apparent efficiency advantage

Symmetric Measures

		Value	Approximate Significance
Nominal by Nominal	Phi	.780	<.001
	Cramer's V	.551	<.001
N of Valid Cases		100	

Table 7

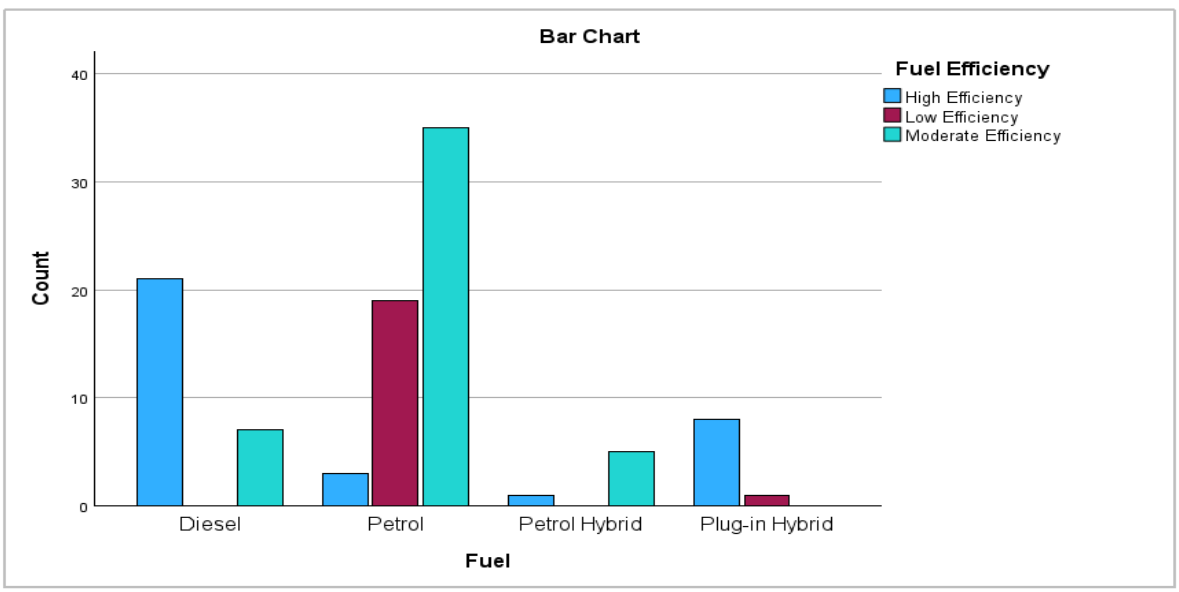


Figure 5

6.4 Analysis of Variance (ANOVA): Comparing Car Prices by Fuel Type

Research Question:

- Do average car prices significantly differ across fuel types (Diesel, Petrol, Petrol Hybrid, and Plug-in Hybrid) in the sample data?

1. Null Hypothesis (H_0): There is no statistically significant difference in the average price of cars across fuel types.
2. Alternative Hypothesis (H_1): There is a statistically significant difference in the average price of cars across fuel types.
3. Statistical Test: One-Way ANOVA
4. Significance Level: 5% significance for a two-tailed test ($\alpha = 0.05$)
5. Calculated Test Statistic: $F(3, 96) = 10.885$, $p < .001$
6. Technical Conclusion: Since $p < 0.05$, the null hypothesis is rejected.
7. Interpretation: The average price of cars is very different depending on the type of fuel they use. Diesel cars are usually cheaper since they are older. On the other hand, hybrid and plug-in hybrid cars cost more because they have newer technology and have greater mileage. Petrol automobiles are in the middle of the price range, which means they are affordable and easy to find. The data shows that the kind of fuel has a big effect on the pricing of secondhand cars. Newer, more environmentally friendly technologies tend to cost more.

ANOVA

Price

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	2453302470.7	3	817767490.23	10.885	<.001
Within Groups	7212009118.7	96	75125094.986		
Total	9665311589.4	99			

Table 8

6.3 CORRELATION MATRIX

	Price	Age of Car	Engine Size	Mileage	Owner count
Price	1				
Age of Car	-0.80	1			
Engine Size	0.24	-0.40	1		
Mileage	-0.78	0.74	0.11	1	
Owner count	-0.69	0.84	-0.02	0.63	1

Table 9

In Table 9, the correlation matrix shows how important factors affect the prices of used VW Golfs. Price has a strong negative association with both Age of Car (r = –0.80) and Mileage (r = –0.78), which means that older cars and cars that have been driven more tend to be worth less. There is a moderate positive association between Age and Owner Count (r = 0.84), which means that older cars are usually sold more often. Engine Size has a weak positive association with Price (r = 0.24), which means that performance affects value, but age and mileage are still the most important factors in depreciation.

Figure 6 backs up this connection: as cars get older, their prices go down in an obvious decreasing trend. The regression line (y = –1713.8x + 27043, R^2 = 0.639) shows that each year older a car is, its value drops by roughly £1,700, and age accounts for nearly 64% of the price difference. This visual evidence supports economic predictions of depreciation, showing that wear and tear over time and how the market sees an item have a big effect on its resale value.

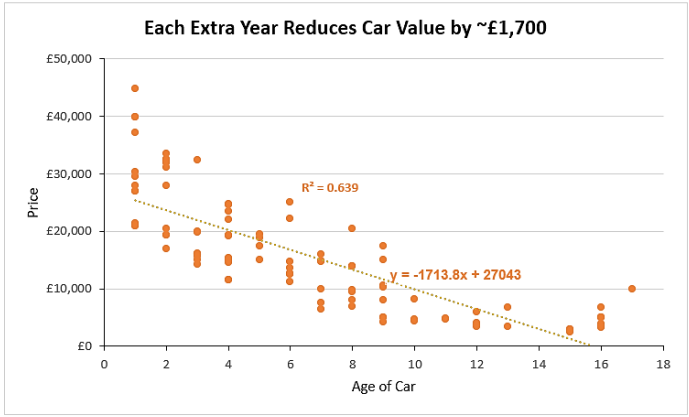


Figure 6

7. Regression Model

7.1 Significant Predictors and Interpretation

We used a multiple linear regression to find the primary factors that affect car prices while keeping the model structure simple. Initial predictors including Color, Presence of Pan-roof, Ownership history, and Fuel Consumption were left out of the research because they didn't offer much to the explanation and made the predictors redundant. The improved model looked at the most important things: mileage, age of the cars, kind of fuel, type of gear, type of drive, and size of the engine.

In Table 10, Adjusted R^2 value of 0.879 indicates that around 88% of the differences in used car prices can be attributed to measurable factors such as age, mileage, engine size, fuel type, gear type, and ownership history. This means that these characteristics collectively provide a strong and reliable explanation of how car prices vary in the market.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.945[a]	.894	.879	3435.090	1.773

a. Predictors: (Constant), EngineSize=2L, Mileage, Fuel=Petrol Hybrid, Fuel=Plug-in Hybrid, EngineSize=1L, DriveType=Front Wheel drive, EngineSize=1.6L, Gear=Manual, Owner, Fuel=Diesel, EngineSize=1.4L, Age of cars

b. Dependent Variable: Price

Table 10

The model can be expressed as:

$$Price = \beta_0 + \beta_1(Mileage) + \beta_2(Age) + \beta_3(Fuel\ Type) + \beta_4(Gear\ Type) + \beta_5(Drive\ Type) + \beta_6(Engine\ Size) + \varepsilon$$

Predictor	B (Unstd.)	Std. Error	β (Std.)	t	Sig.	VIF	Interpretation
Constant	31,990.49	1,336.69		23.93	<.001		Baseline car price
Mileage	-0.08	0.01	-0.34	-5.86	<.001	2.68	Higher mileage lowers value
Age of cars	-933.89	170.48	-0.45	-5.48	<.001	5.42	Older cars priced lower
Owner	333.24	310.25	0.07	1.07	0.286	3.51	No significant impact
Fuel = Diesel	-3,918.73	1,172.40	-0.18	-3.34	0.001	2.35	Diesel cars priced lower
Fuel = Petrol Hybrid	-2,445.69	1,794.47	-0.06	-1.36	0.176	1.54	Hybrid effect unclear
Fuel = Plug-in Hybrid	2,124.97	1,579.38	0.06	1.35	0.182	1.73	Limited data available
Gear = Manual	-3,124.13	905.56	-0.16	-3.45	<.001	1.73	Manuals cheaper than autos
Drive = Front wheel	-5,086.40	1,325.46	-0.18	-3.84	<.001	1.79	FWD cars cheaper
Engine size = 1.4L	-2,561.46	1,558.89	-0.1	-1.64	0.104	2.77	Minor price difference
Engine size = 1.6L	976.39	1,593.70	0.03	0.61	0.542	2.59	No significant effect
Engine size = 1L	-3,523.99	1,561.63	-0.09	-2.26	0.027	1.35	Small engines reduce value
Engine size = 2L	2,635.92	1,149.16	0.13	2.29	0.024	2.66	Larger engines increase value

Table 11

The price of Golf is significantly negatively correlated with both its mileage and its age ($p < .001$). Similarly, type of fuel also affects the price: Diesel automobiles cost less than cars in other categories as they are oldest. The same tendency can be seen in transmission type: manual cars are worth less than automatic cars because people want convenience and comfort.

Engine size has a subtle effect on mechanical properties. Smaller engines (1.0L–1.4L) are linked to lower pricing, whereas bigger engines (2.0L) are linked to higher costs. Larger engines and stronger performance increase a car's perceived value. Drive Type also influences pricing. For example, front-wheel drive automobiles are usually less expensive than four-wheel drive cars. This could be because of differences in traction, performance, and vehicle category.

The VIF values are still much below 5, which shows that multicollinearity is not a problem. The results show that the most important factors that affect automobile prices in the dataset are the condition of the vehicle (age and mileage), the mechanical configuration (engine and drive type), and the transmission preference.

7.2 Price Calculator

Price = B0 + (B1 * Mileage) + (B2 * Age) + (B3 * Owner) + (B4 * Fuel Type) + (B5 * Gear Type) + (B6 * Drive Type) + (B7 * Engine Size)

Where:

- B_0 = Constant (intercept)
- Each B_i = Coefficient for that predictor
- Dummy variables (Fuel, Gear, Drive, Engine Size) take the value 1 if applicable and 0 otherwise

Example:

For a 2021 petrol automatic (non-hybrid) 1.5L, front-wheel drive car with:

- Mileage = 45,000
- Age = 4
- Owner = 1

Since this is a petrol (baseline fuel) and automatic (baseline gear) car, all dummy variables are 0 except:

- Drive = Front wheel → 1

Price = 31,990.49 + (−0.08 × 45,000) + (−933.89 × 4) + (333.24 × 1) + (0) + (0) + (0) + (0) + (−5,086.40 × 1) + (0)

Price = 31,990.49 − 3,600 − 3,735.56 + 333.24 − 5,086.40

Predicted Price = **£19,901.77**

Therefore, this 2021 petrol automatic (45,000 miles) would be valued at approximately £19,900 according to the model.

7.2 Regression Diagnostics and Assumption Testing

The regression diagnostics were examined to assess the key assumptions of normality, linearity, and homoscedasticity (Statistics, 2025). Overall, the model demonstrates a good fit, with most assumptions reasonably met. Minor deviations appear mainly at the price extremes, reflecting both genuine market variation and sample imbalance within the dataset.

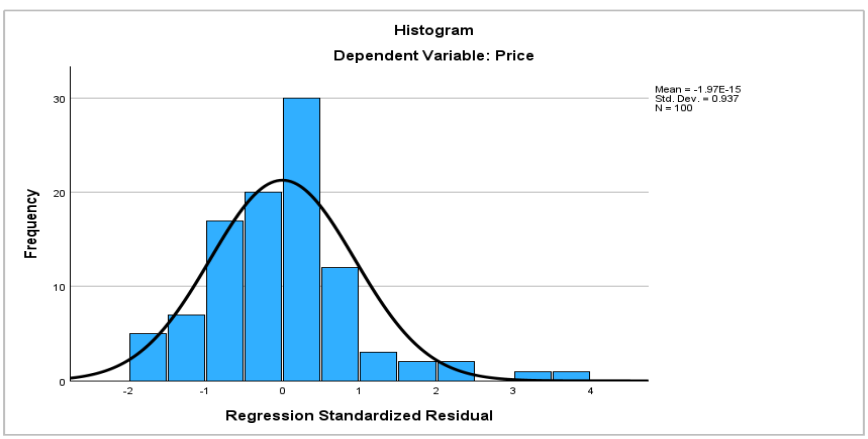


Figure 7

The histogram shows that most used Volkswagen Golf cars in Birmingham are priced within a typical range, forming a near bell-shaped curve. This means that the majority of listings sit close to the expected mid-market value, with few extreme prices. The slight right skew appears because a handful of newer, high-spec cars such as the 2025 Plug-in Hybrid Automatic priced at £44,845 and several 2.0L petrol automatics ranging from £37,000 to £40,000 — sell for much higher than the average £10K–£20K range.

Additionally, the relatively small number of hybrids in the sample may have amplified this effect, as their higher prices stand out more strongly against the larger share of standard petrol and diesel cars.

The points on the P–P plot is indicating that the residuals are roughly normally distributed. There are small differences at the lower and upper ends, specifically for older diesel cars (2008–2012, priced below £6,000) and newer hybrid or plug-in hybrid cars from 2023–2025 (priced above £30,000).

This means that the model is very accurate for mid-range cars (costing between £10,000 and £20,000), but not quite as accurate for the cheapest and most expensive cars. In real life, the depreciation patterns for budget diesel models and new hybrid variants don't follow the same curve, so the regression doesn't fit these groups as well as it should.

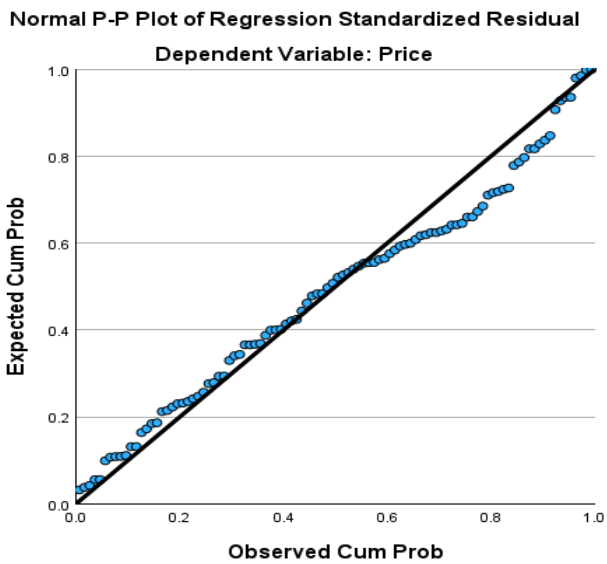


Figure 8

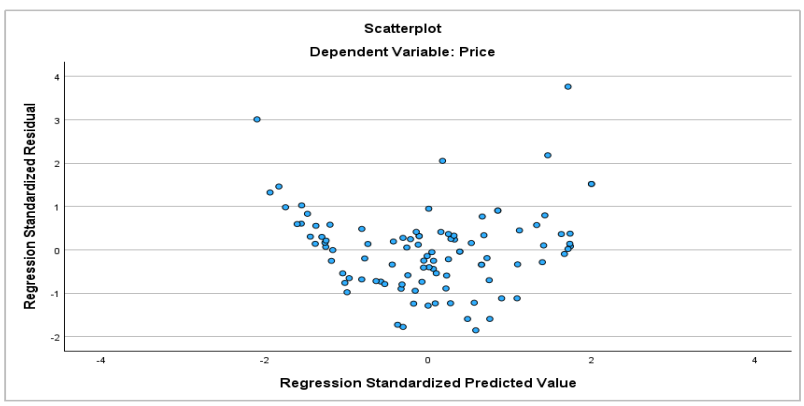


Figure 9

The scatterplot tests whether the model's errors are randomly distributed, and overall, the residuals appear fairly balanced around zero, indicating that the model does not consistently overestimate or underestimate prices. However, a slight U-shaped pattern can be observed, where the residuals widen at both the lower and higher ends of the price range. This occurs because vehicle depreciation does not follow a perfectly straight line. Newer, high-spec models such as plug-in hybrids and 2.0L petrol automatics lose value more sharply in their first few years, while older diesel manuals from around 2008 to 2012 have already reached a low-price floor, meaning their values no longer decrease substantially. Additionally, the uneven distribution of fuel types within the dataset contributes to this pattern. Out of 100 listings, only around 15 were hybrid or plug-in hybrid models, while most were petrol or diesel. Including more hybrid cars might have balanced the pattern, but even when plug-in and petrol hybrids were combined for analysis, the results remained statistically insignificant.

8. Recommendation

- Focus on petrol automatics worth between £10,000 and £20,000: About 80% of listings are below £20,000, and petrol automatics make up most of this range. ANOVA shows that there are big price disparities between fuel types. For example, automatics cost about £8,000 more than manuals. To maximize turnover and profit, stock mostly mid-priced petrol automatics and limit low-end diesels.
- Chi-square shows that there is a substantial link between fuel and efficiency. Diesels are better for low-cost, high-efficiency commuting, whereas hybrids are the eco-premium category. Instead of seeing all fuels as replacements, set prices and promotions based on these functions.
- Regression demonstrates that 4WD (+£5K), automatics (+£3.1K), and 2.0L engines (+£2.6K) all have high premiums. Black models are available in both low- and high-price ranges, but they sell faster in premium trims, which shows that buyers prefer them. To get people in the premium category to buy, list and show high-end black automatics as "hero cars."

9. Appendix

- **SPSS Output Tables:** key statistical test outputs generated using IBM SPSS Statistics. These include results from the One-Sample T-Test, Chi-Square Test, Correlation Matrix, ANOVA, and Regression Analysis. Please see few outputs below –

Oneway

Descriptives

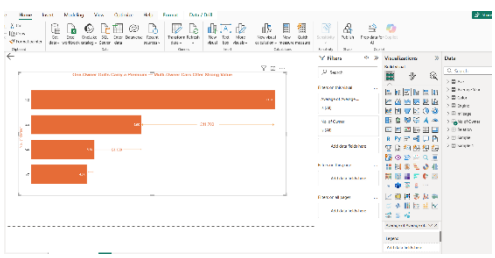
Price	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean Lower Bound	Upper Bound	Minimum	Maximum
Diesel	28	8178.29	5281.238	998.060	6130.44	10226.14	2990	24621
Petrol	57	18445.37	9728.931	1288.628	15863.94	21026.80	2395	39814
Petrol Hybrid	6	17949.83	3315.428	1353.518	14470.51	21429.16	11202	19499
Plug-in Hybrid	9	22476.89	11744.473	3914.824	13449.29	31504.49	7990	44845
Total	100	15903.69	9880.760	988.076	13943.13	17864.25	2395	44845

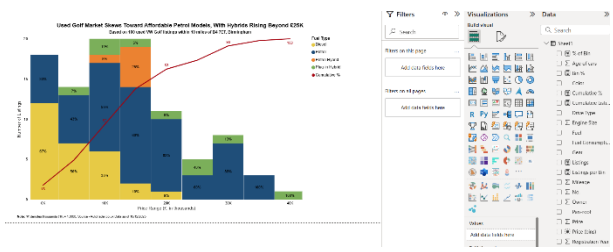
Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	VIF
1	(Constant)	31990.489	1336.694		23.933	<.001		
	Mileage	-.080	.014	-.335	-5.858	<.001	.374	2.676
	Age of cars	-933.893	170.479	-.445	-5.478	<.001	.185	5.415
	Owner	333.237	310.251	.070	1.074	.286	.285	3.505
	Fuel=Diesel	-3918.730	1172.395	-.179	-3.342	.001	.426	2.348
	Fuel=Petrol Hybrid	-2445.692	1784.471	-.059	-1.363	.176	.650	1.539
	Fuel=Plug-in Hybrid	2124.971	1579.384	.062	1.345	.182	.578	1.731
	Gear=Manual	-3124.125	905.557	-.158	-3.450	<.001	.579	1.726
	DriveType=Front Wheel drive	-5086.398	1325.462	-.180	-3.837	<.001	.558	1.793
	EngineSize=1.4L	-2561.464	1558.893	-.096	-1.643	.104	.361	2.768
	EngineSize=1.6L	976.389	1593.701	.034	.613	.542	.386	2.592
	EngineSize=1L	-3523.987	1561.626	-.091	-2.257	.027	.743	1.345
	EngineSize=2L	2635.917	1149.164	.131	2.294	.024	.376	2.662

a. Dependent Variable: Price

- **Power BI Visuals:** Power BI tool was used to illustrate the patterns, trends, and statistical insights presented in the analysis section.





- **Use of AI Assistance:** This report made limited use of ChatGPT (GPT-5) as a language refinement and paraphrasing tool. Here are few samples prompt that we have been used

  - Compress this paragraph and use academic tone
  - Please note that i can't write long passage, i have page limitations, so make it short precise and on point but don't miss out key takeaways from the paragraph that has been provided.
  - Price difference was kept with equal bins to follow tufte's idea of honesty. Joints four charts close together to make it easily comparable. Note has been given to follow Gestalt rules like "£7,724 less than newer one". Silver color cars were purposefully visualized as pink as we can't detect silver on white background. bright color and notes were used purposefully even though it violated data-ink ratio. Please paraphrase this passage using academic tone and keeping facts as it is.

- Use of Excel: Microsoft Excel was used for data storage, cleaning, and preliminary analysis. It supported the creation of tables, calculation of descriptive statistics, and exploration of initial relationships between variables before conducting detailed statistical tests in SPSS and Power BI.

10. References

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